Sunflower Shelling Machine Design Documentation

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Subject: Sunflower Shelling Machine Design Process Documentation

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2.1 Executive Summary

This project developed a sunflower shelling machine that cracks open sunflower seeds and separates the shells from the kernels. The engineers' design objective was to maximize the machine's shelling success rate within the 3-minute performance window, where performance is measured as kernels produced per minute, minus a penalty for waste/unprocessed seeds. The engineers developed a two-subsystem machine: (1) a roller cracking mechanism and (2) a multi-level filtration system. For cracking, the engineers' final configuration pairs one horizontal-bump roller with one vertical-bump roller, which, during testing, showed improved performance compared to homogenous roller designs. For separation, a two-stage filter system sized through sampling and experimentation was implemented. To prevent clogging and improve usability, both filters are semicircular and include handles for twisting, helping dislodge trapped material during operation. Finally, the design meets key challenge constraints: the machine is solely human-powered, processes all seeds from a single hopper, and is designed to separate kernels from shells to facilitate counting within the demo window. During testing on both trials, the team managed to crush 1 seed, with 14 seeds wasted, resulting in a yield of -2 seeds per minute.

2.2 Approach Description

Ideation Techniques Used

Sunflower seeds can be difficult to enjoy, as many people have difficulty cracking them. With this issue in mind, a machine was created. The engineers came up with different ideas to benefit different stakeholders. Some stakeholders the engineers brainstormed include elderly and young children who enjoy eating sunflower seeds but lack the strength to crack the seeds, companies that benefit from separating the shells from the kernel (in the lens of effectiveness), and normal people who simply want to enjoy this snack without having to struggle with crushing the seeds manually.

To overcome the design barrier, different ideas were drafted. After a thorough brainstorming session, a variety of ideas emerged. These ideas were drawn on a whiteboard, and the team commented on benefits and challenges that each solution would have. There was some dissonance on what method to pick.

The potential methods were separated in terms of speed, aesthetics, and strength. The three main ideas consisted of a seed-splitter tool, a crusher plate, and 2 sets of rollers with a crank.

Idea Selection and Refinement Techniques

The seed-splitter tool would have a similar structure to a lemon juicer or a nut cracker. This tool would consist of a small hole in the middle of a pair of pliers that the engineers would 3D print. This design was eventually discarded as the team decided to move towards something more aesthetic (or machine-like) and with a higher processing speed.

The crusher plate simply consists of a wooden plate that would press the seeds with a given orientation to split them open. The team's take on this design (and the reason why it was discarded) is that it was blunt and unattractive. This design, however, improved the speed and strength of the machine, as more seeds could get processed at the same time while serving as a force amplifier.

Lastly, the design that the team decided to implement was the 2 sets of rollers with a crank. This design values aesthetic and strength as the lever works as a force amplifier while having a clean look. After consideration, the team decided to move on and implement this design.

The engineers' design objective was to maximize the machine's shelling success rate within the 3-minute performance window, where performance is measured as kernels produced per minute, minus a penalty for waste/unprocessed seeds. To align with the scoring definition of a "clean kernel" (fully removed, undamaged, and free of adhering material), the system prioritizes cracking shells without pulverizing the seed, then separating the outputs cleanly for easy counting.

Metrics used for this design were the average width and length of a sunflower seed, measured with a caliper. These metrics were key to the design because they determined how wide the ridges of the rollers were going to be, as well as how far apart the rollers were separated from each other.

Voting, Value Matrix, and How-Why Diagram

After the initial brainstorming session, the team used voting as a preliminary filtering method to narrow down the large number of ideas generated. Each team member voted on concepts based on perceived feasibility, performance potential, and alignment with stakeholder needs. This process allowed the team to quickly eliminate ideas that did not meet basic constraints while ensuring that all members had input in the decision-making process.

Following voting, a value matrix (see Table 1) was used to quantitatively compare the remaining design concepts. Criteria such as processing speed, kernel damage risk, aesthetic appeal, manufacturability, and mechanical advantage were identified and weighted based on their importance to the project objectives. Each concept was then scored against these criteria, allowing the team to objectively evaluate trade-offs between designs. The value matrix highlighted that the roller-based cracking mechanism provided the best balance between performance, aesthetics, and reliability, leading to its selection for further development. Table 1 and Figure 1 show the Value Matrix and How-Why Diagram (key in decision-making).

Table 1: Value Matrix

Criteria	Weight	Seed-Splitter Tool	Crusher Plate	Dual-Roller System
Processing Speed	0.25	2	4	5
Kernel Damage Risk	0.2	3	2	4
Manufacturability	0.15	4	3	4
Professional Appearance	0.15	2	1	5
Mechanical Advantage	0.15	3	4	5
Reliability / Consistency	0.1	3	3	4
Weighted Total Score	1	2.75	2.95	4.65

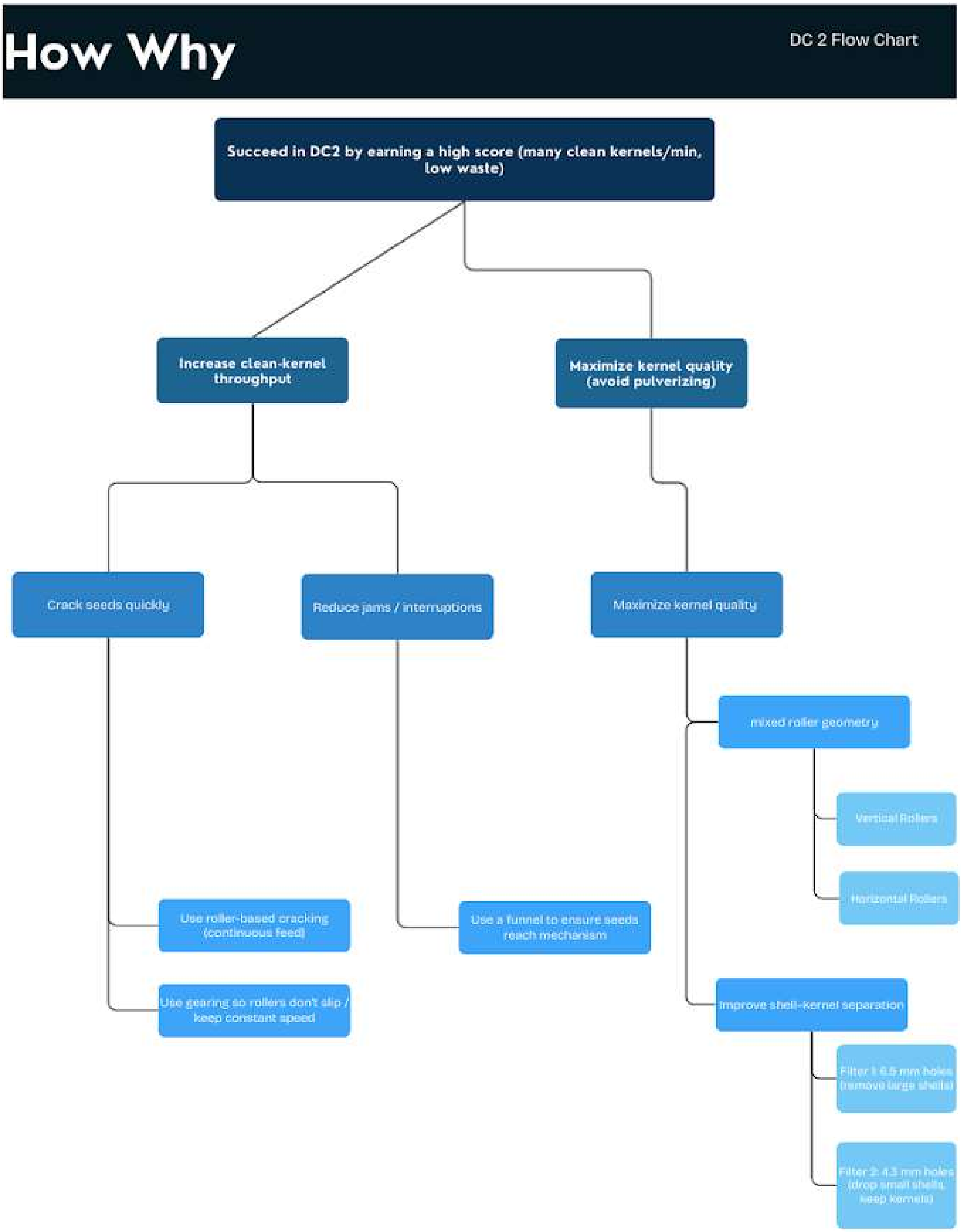

Figure 1: How-Why Diagram

Methods (Prototyping, CAD, and Manufacturing)

Methods for Rollers

To brainstorm the design of the cracking apparatus, the team began by looking at existing solutions. The most efficient ones seen in the research utilized systems with two rollers connected by gears that rolled the same direction and cracked seeds in between them. The team's goal was to replicate this type of design with 3D printed parts, but add some variations.

Testing was done with rough first drafts to see which type of rollers would work best for separating seeds. Materials such as cardboard, wood, and spare plastic parts were used. At the end of this testing period, the engineers decided that a bumpy surface worked best, and decided to use this for their 3D model made in SolidWorks, as shown in Figure 2 (more details in Appendix B3 and B4). Measurements were made based on the thickness of sunflower seeds measured with calipers.

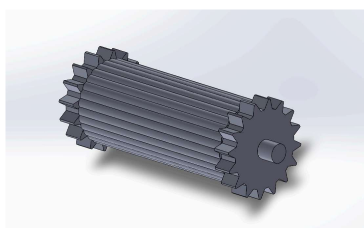

Figure 2: Roller With Horizontal Bumps

Upon printing the rollers with the horizontal bumps and attempting to crush seeds with them, the engineers noticed that although the shells were being cracked successfully, many of the seeds inside were being smashed as well. Initial solutions consisted of a wider gap between the rollers and smaller bumps on the roller. These designs also led to failure, as the wider gap simply made the seeds fall through completely, and the smaller bumps crushed the seeds entirely. The engineers concluded that the space between rollers wasn't the issue, but rather the way the roller was designed. The current rollers applied the same force across the entire shell, which crushed both the shell and the seed inside. To solve this, they designed a roller with vertical bumps, shown in Figure 3 (more details in Appendix B5), which would apply a force at the top and bottom of the shell, leaving the seed unharmed.

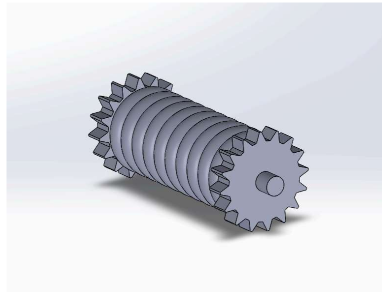

Figure 3: Roller With Vertical Bumps

When they attempted to use both rollers with vertical bumps, the engineers saw that many of the seeds would not get cracked at all, and simply slipped through the gaps between the bumps. To solve this issue, the team decided to use one vertical roller and one horizontal roller. This successfully balanced out the distributed force of the horizontal bumps and the point force applied by the vertical rollers, leading to an optimized cracking system.

Methods for Filter

Another important factor in the design was separating the shells from the kernels so that the user has an easier time using the machine. For separation, the engineers tested fan-based and single-filter concepts but encountered inconsistent separation and clogging.

When designing the filter, the engineers first brainstormed and prototyped various ideas. Initial prototypes consisted of a fan controlled by a gear mechanism that would blow the shells away. However, the issue with this was that some seeds were too light and would blow away as well, and some heavy seeds wouldn't move.

Another idea that was tested was a single filter that would allow for seeds to pass but keep shells behind. Although functional at first, as more seeds were added to the machine, the filter ended up clogging. The engineers sought to improve upon this idea by creating multiple filters to avoid clogging. They started by experimenting with different-sized filter holes, and ultimately chose holes of 4.3mm and 6.5mm. These sizes were picked after sampling multiple shells and observing their performance with different hole diameters. These filters were made from 3D printed PLA material since it was easy to work with and lightweight.

Furthermore, to make sure seeds and shells fell through the filter, the engineers designed the filters in a semicircular shape and added handles to allow for twisting. This helped dislodge any seeds or shells that got stuck.

The first filter had holes that were 6.5mm in diameter and served to keep large shells but allow smaller shells and seeds to pass, as seen in Figure 4 (more details in Appendix B1). This helped avoid clogging in the next filter.

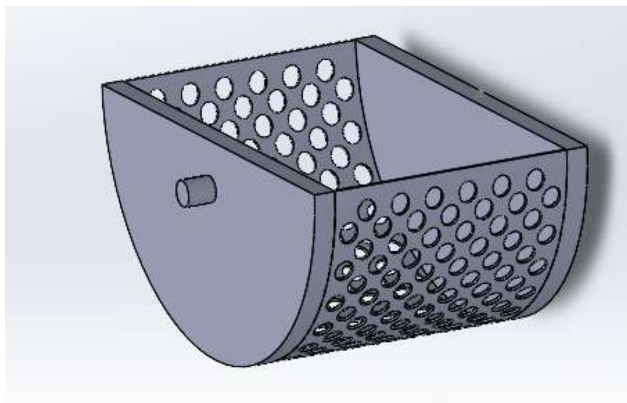

Figure 4: First Filter

The next filter had 4.3mm holes that allowed for small shells to pass but kept seeds inside, as shown in Figure 5 (more details in Appendix B2).

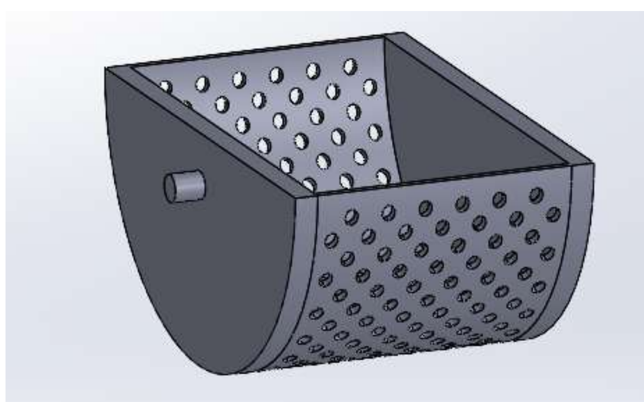

Figure 5: Second Filter

Method for Funnel

In order to make the cracking device fully automated, the team needed a hopper to feed seeds into the rollers. Only one solution was brought up, as it was a functional solution, and the hopper was not integral enough to the functionality of the project to brainstorm many solutions. The team decided to use a funnel to put the seeds in the crusher, and to use a strip at the bottom to stop the flow of seeds when they wanted to. Due to time constraints, the engineers opted to use cardboard for the funnel, as it is easy to cut, manipulate, and construct, along with it being lightweight. The funnel was made in a way that it was wide enough to let seeds through easily, but also thin enough so the engineers could make sure that all the seeds were falling into the rollers. A drawing of the funnel is shown in Figure 6.

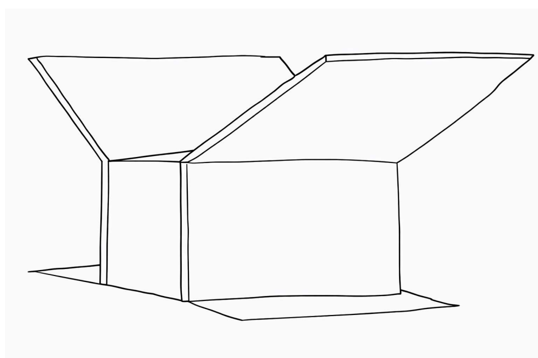

Figure 6: Seed Hopper

Methods for Body

The general structure of the machine was made of wood. This method was preferred, as the building time was quicker and more convenient than that of 3D Printing. Additionally, wood was used because it is a stronger material than PLA, which means that it is better suited for a machine that will withstand a great amount of force due to torque. Figure 7 shows the full machine with its 3 subsections. In Appendix A, there are 2 alternative views of the model to see the overall shape (more details in Appendix A1). In order to accomplish the desired structure, different tools were used. For the holes, a drill press was used so that it has a cleaner look than a manual drill. For straight cuts, multiple tools were used depending on the necessary cut, especially a bandsaw and a compound miter saw. For some of the wood pieces, the team filed down areas inside the machine to allow for the filtration devices to run smoothly.

Figure 7: Complete Machine without Funnel

Final Integrated Systems, Results, and Conclusion

The engineers combined the best of their brainstormed ideas and developed a two-subsystem machine featuring a roller cracking mechanism in a PLA box (more details in Appendix B6) and a multi-level filtration system. This design met the key challenge constraints of being solely human-powered, processing all seeds from a single hopper, and separating kernels from shells for easy counting.

During the presentation, there were two test trials, both lasting 3 minutes. On both attempts, the engineer's design managed to successfully crush one seed, while leaving 14 seeds wasted. This gave an overall yield of -2 seeds per minute.

The engineers came into this project looking to crush shells from sunflower seeds without harming the kernel. They came up with a solution that incorporated multiple different parts consisting of: a triangular cardboard funnel, a crusher consisting of a vertical and horizontal roller, and a two filter system used to separate shells from seeds. This met the constraints of being solely human-powered, processing seeds from a hopper, and separating the shell from the seed.

2.3 3D Machine Sketch

Below are sketches of the machine. Figures 8-11 show specific parts of the seed cracking machine.

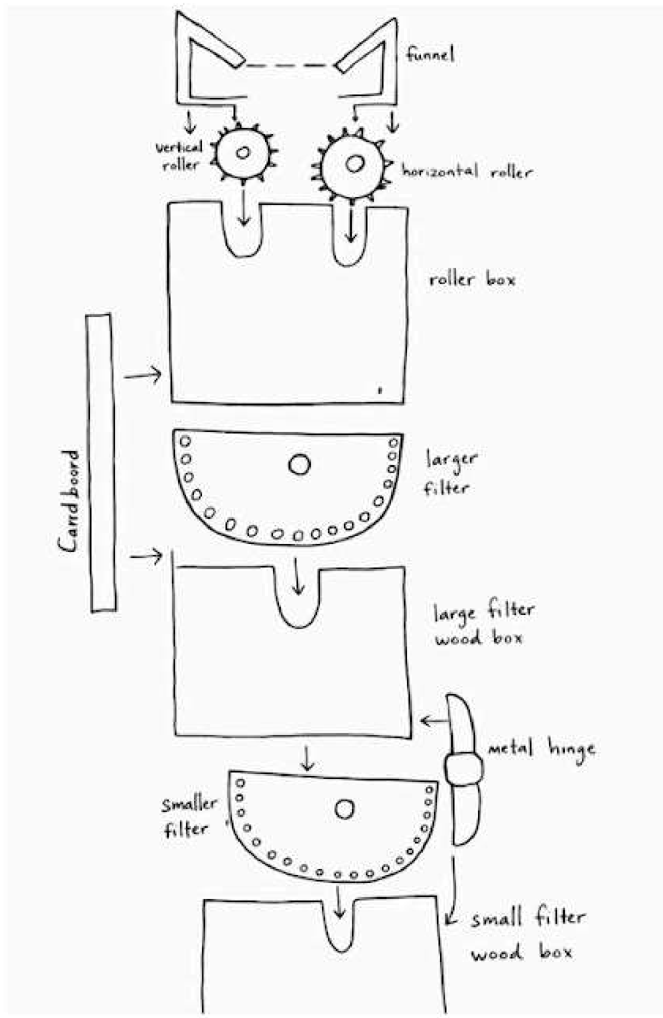

Figure 8: Exploded View

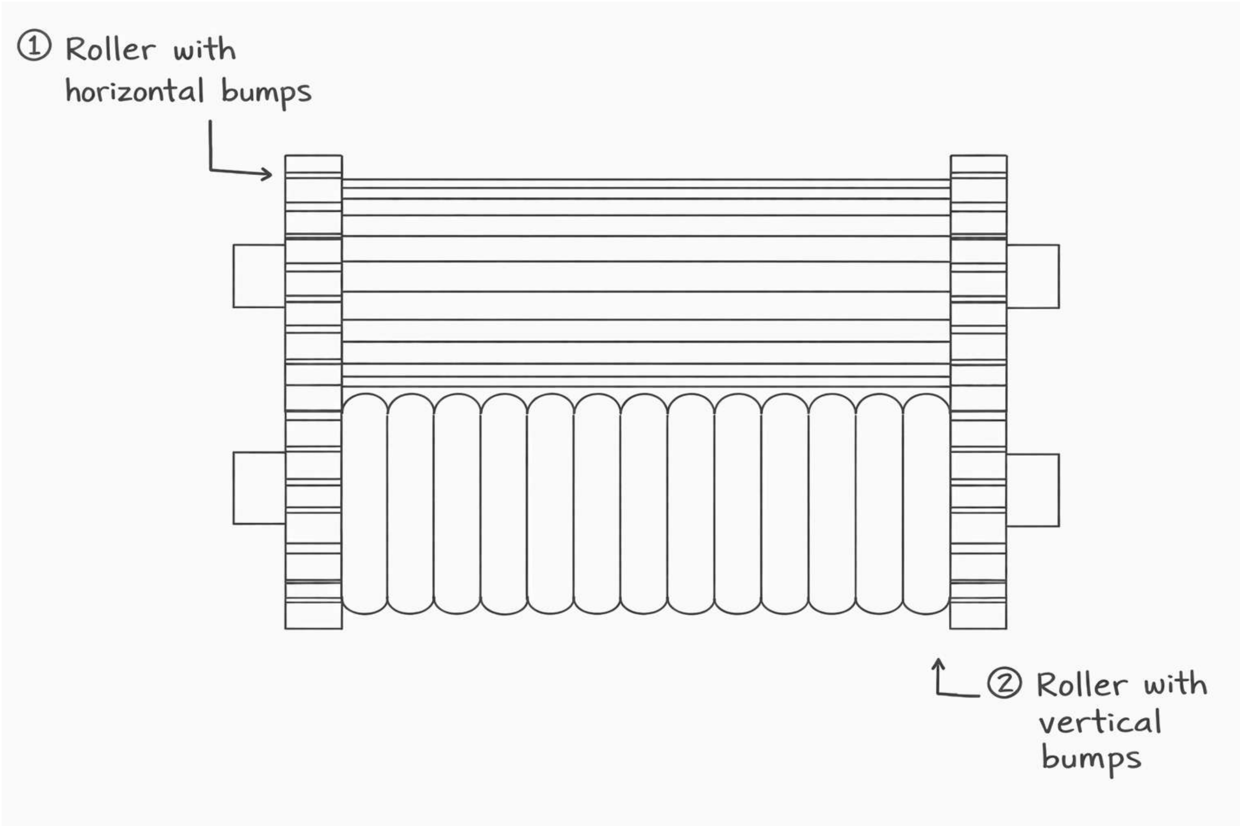

Figure 9: Sketch of Roller Component

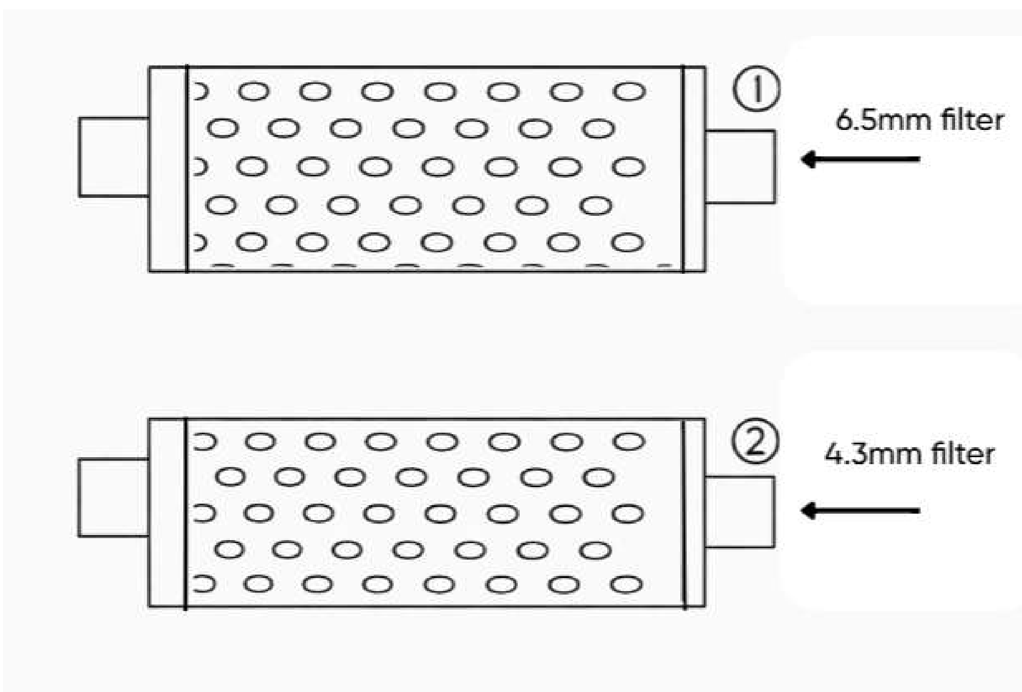

Figure 10: Sketch of Filter Component

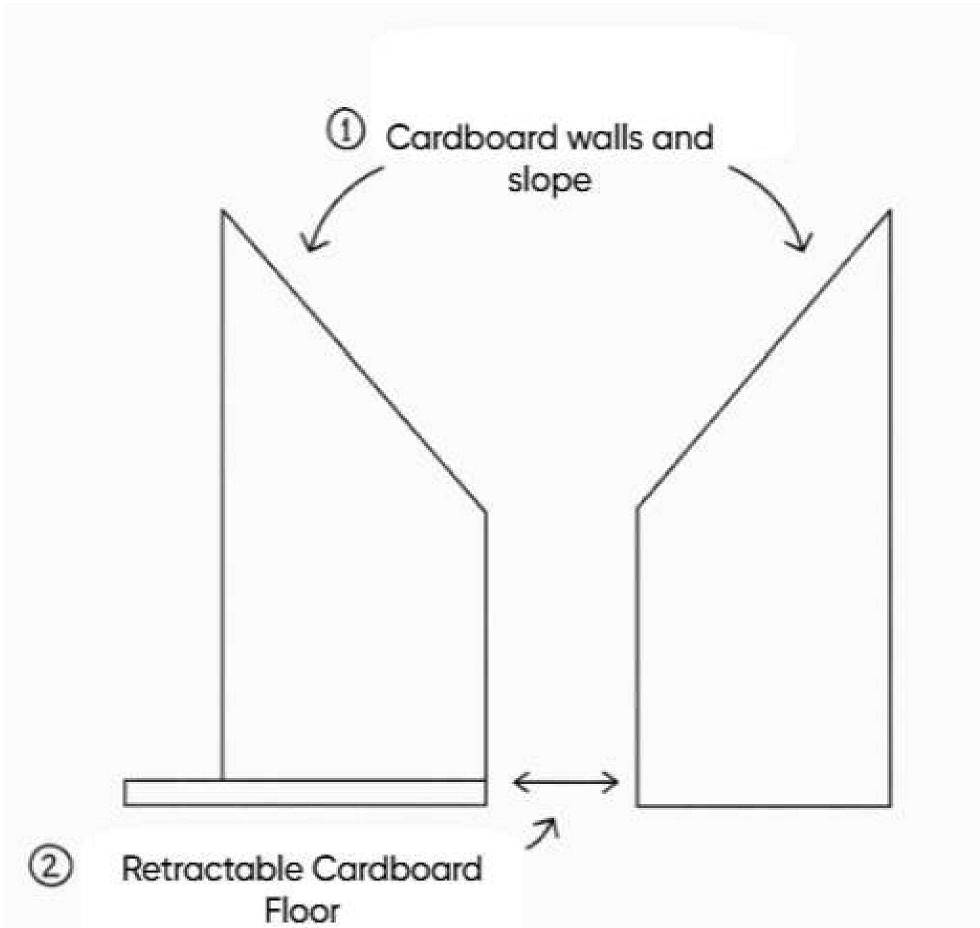

Figure 11: Sketch of Funnel Component

2.4 Team Reflection

During DC2, we struggled to crush seeds correctly during the presentation, even though it was working beforehand. This could be due to how the seeds were placed into the machine. Before the presentation, we dropped the seeds slowly and were calm while crushing. However, on presentation day, we were stressed and tried to rush the crushing process, leading to mistakes. If we were to do the project again, we would try to simulate the presentation by timing tests. This could help prepare us for when we go to present.

Furthermore, our lack of knowledge of the project led to time being wasted, which could have been used for other parts of the project. We thought that the presentation also graded whether the seeds and shells were in separate piles. However, we were only graded on whether the seed was removed from the shell. If we were to repeat this design challenge, we would start by properly going over the rules and figuring out a plan early.

A major challenge we faced during the project was coming up with ways to crush the shells without destroying the seed. We tried multiple variations of rollers, but in the end, we could not come up with a consistent crusher. We tried to address this by prototyping multiple different crushers. In the end, we chose the crusher that had the most success.

On the other hand, we were able to effectively communicate and plan out the timeline for the project. After completing DC1, we realized the importance of starting early and began prototyping immediately. We scheduled multiple meetings and calls and spent time working on assigned parts. There was collaboration throughout all parts of the project, and members updated each other consistently on their progress.

Throughout the course of this design challenge, our team started to become more comfortable with SolidWorks and were able to make prototypes much quicker. We also became more comfortable with the tools in the foundry. Previously, none of us had really messed around with different tools, but since we could not just print everything due to time constraints, we had to manually make components such as the casing. Our team is leaving this project with a stronger understanding of SolidWorks and technical skills that will aid us during future design challenges.

Appendix

Appendix A. Alternative Views of Machine

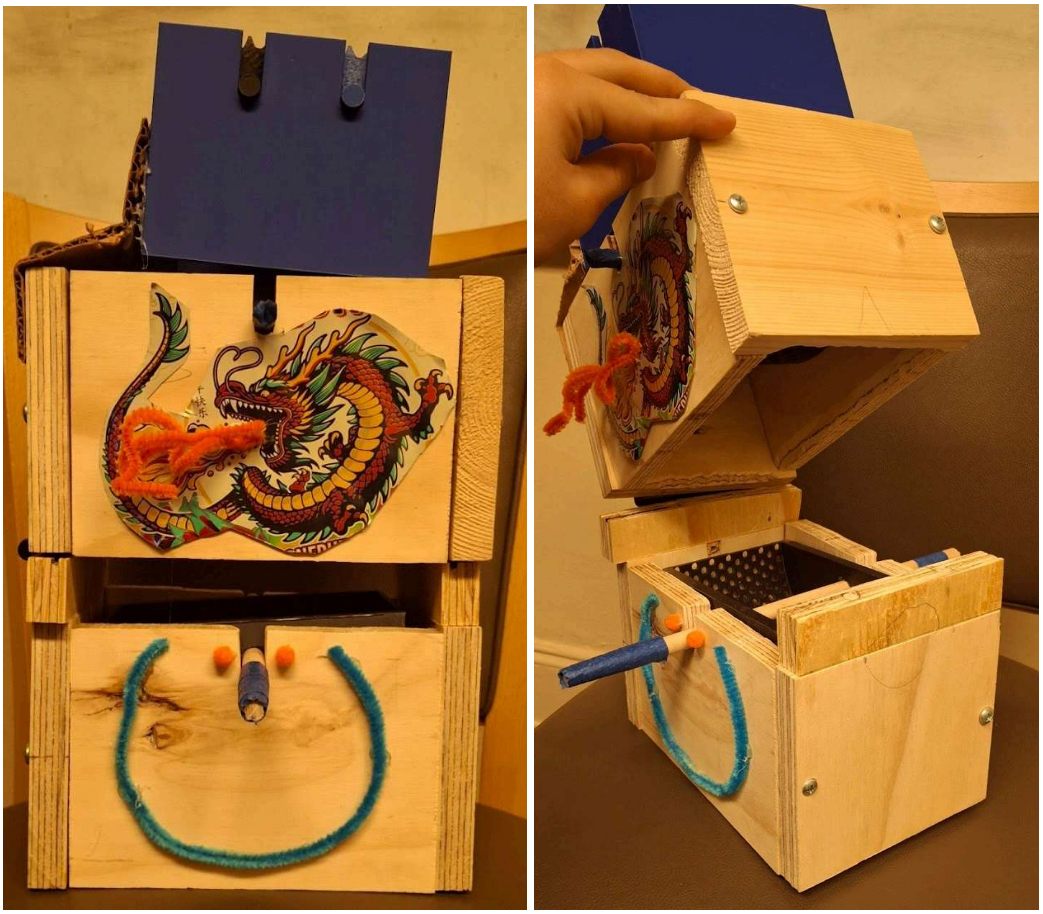

Figure A1. Front and Side View of Final Design

Appendix B. Dimensioned Part Drawings

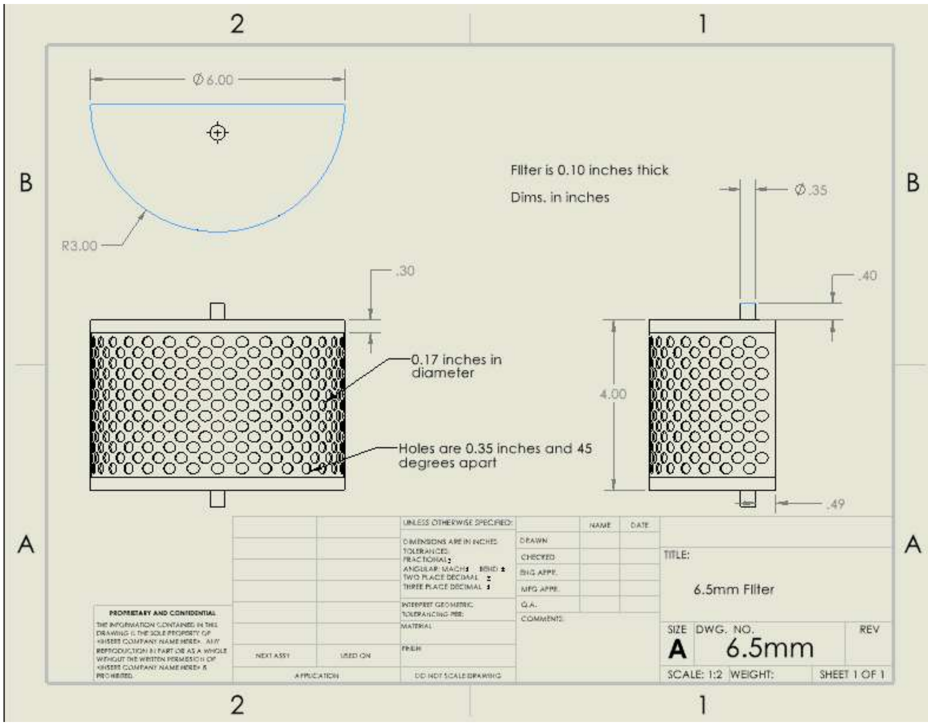

Figure B1. Dimensions of 6.5mm Filter

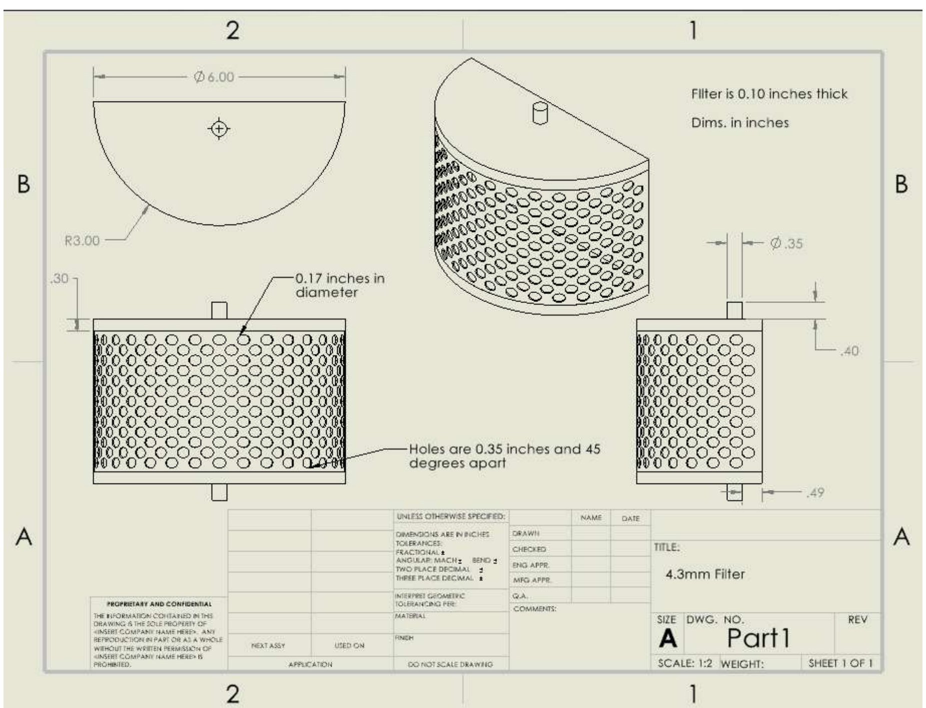

Figure B2. Dimensions of 4.3mm Filter

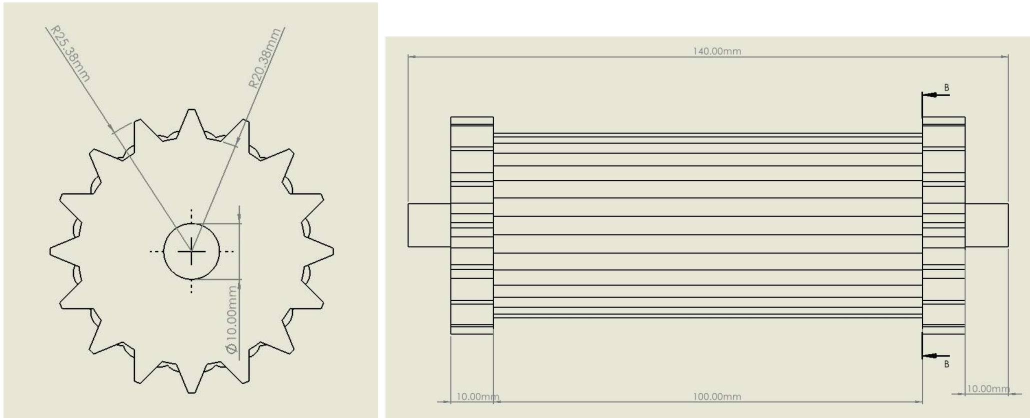

Figure B3. Dimensioned Front and Side View of Roller With Horizontal Bumps

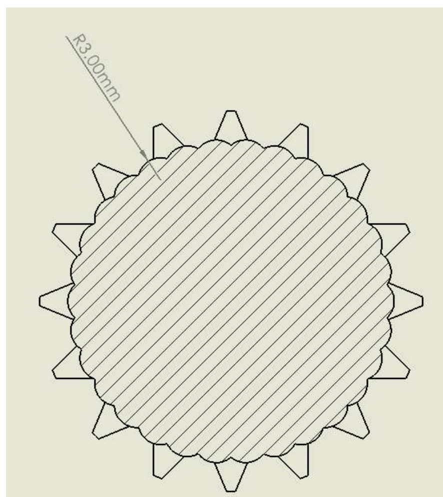

Figure B4. Dimensioned Cross-Section View of Roller With Horizontal Bumps

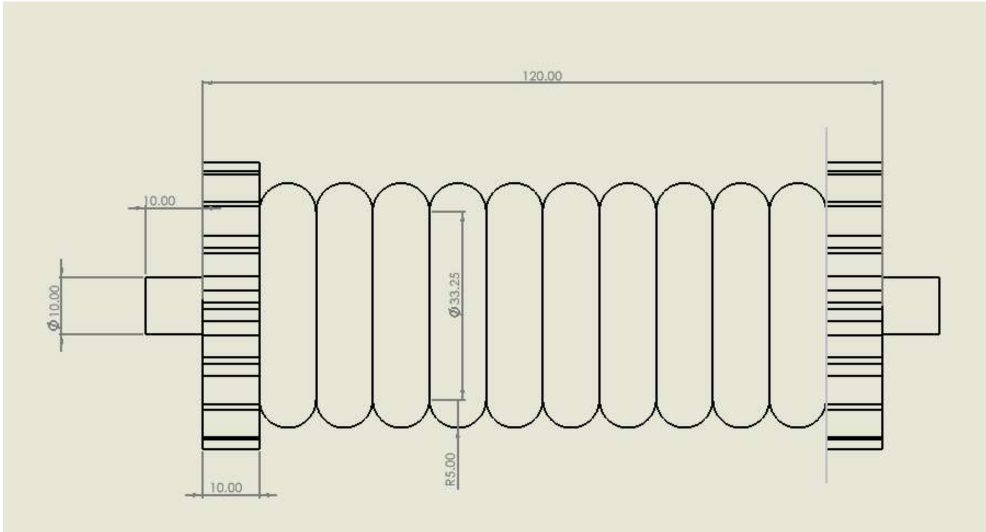

Figure B5. Dimensioned Right Side View of Roller With Vertical Bumps

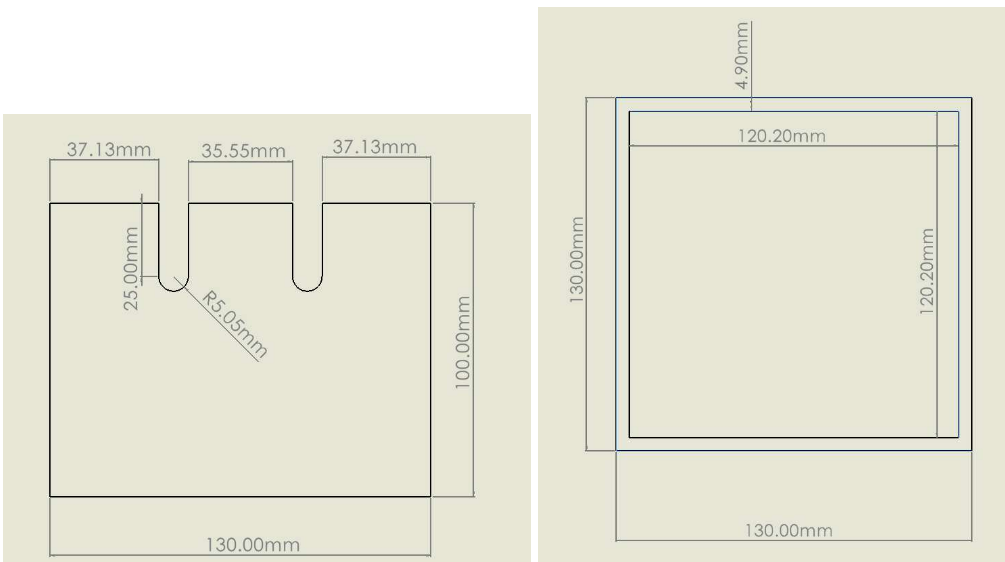

Figure B6. Dimensioned Right Side and Top View of Roller Box